

LESSON 2-5 • SESSION 2 • PRACTICE • FORM B

Comparing Variability

Learning target – I can compare the variability of two data sets using range and IQR. • 6.SP.B.5c

REMEMBER

- **More consistent** = less spread out = smaller range or smaller IQR.
- Compare the **same** measure for both groups. Equal ranges can hide different IQRs.
- With one far-out value, trust the **IQR** more than the range.

WORKED EXAMPLE

Class R: range 18, IQR 12. Class S: range 35, IQR 25.

Step 1 Ranges: $18 < 35$.**Step 2** IQRs: $12 < 25$.**Step 3** Both measures are smaller for R.**Answer** Class R is more consistent**WORDS** consistent (*constante*) • range (*rango*) • IQR (*rango intercuartil*)

PRACTICE

show your work • label units

1. **Two Teams.** Team J scores: 64, 78, 71, 83, 67, 80, 74. Team K scores: 55, 92, 70, 88, 60, 81, 66. Complete the table.

Hint Order each list on its own first.

	Range	Q1	Q3	IQR
Team J				
Team K				

More consistent _____

2. **Vending Machines.** Machine 1 daily sales: 28, 45, 33, 52, 36, 48, 31. Machine 2: 38, 42, 44, 40, 47, 43, 49.

a. Range 1: ____ b. IQR 1: ____ c. Range 2: ____ d. IQR 2: ____

3. **Aaliyah's Claim.** Sets V and W both have a range of 50. V has $Q1 = 20$ and $Q3 = 45$; W has $Q1 = 30$ and $Q3 = 36$. Aaliyah says they vary the same amount.

Hint Equal ranges only match the extremes.

☐ Right ☐ Not right

IQRs _____

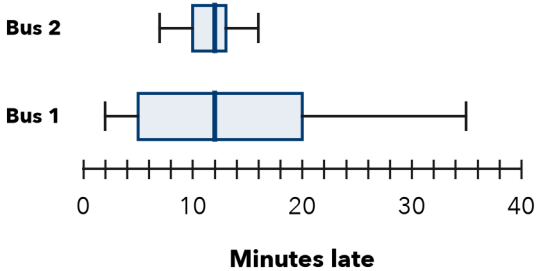
4. **Mateo's Claim.** Sets T and U both have an IQR of 10. T runs from 40 to 62; U runs from 12 to 95. Mateo says they are equally spread out.

Hint Equal IQRs only match the middle halves.

☐ Right ☐ Not right

Ranges _____

5. **Choosing a Measure.** A list of monthly phone bills has one bill far above all the others. Which measure of spread fits better, the range or the IQR? Explain.

- 6. Different Stories.** Set M has range 36 and IQR 20. Set N has range 50 and IQR 5.
Hint A range much bigger than the IQR points to an extreme value.
- a. Wider middle half: ____ b. Likely has one unusual value: ____
- 7. Tablet Batteries.** Tablet 1 hours: 10, 15, 12, 19, 17, 21, 8. Tablet 2 hours: 13, 15, 14, 17, 12, 18, 16.
- a. Range 1: ____ b. IQR 1: ____ c. Range 2: ____ d. IQR 2: ____
- 8. Choosing a Bus.** The box plots show how many minutes late each bus was.
- a. IQR of Bus 1: ____
 b. IQR of Bus 2: ____
 c. More predictable: ____
 d. Bus 1 is sometimes almost on time. Why might you still choose Bus 2?
- 

FIND THE ERROR *explain the mistake, then fix it*

- 9. Nadia's Work.** Nadia wrote: "Set A has the greater IQR, so Set A must have the greater range too." Explain the mistake.
- _____
- _____

TEST PRACTICE *choose the best answer*

- 10.** Set X has range 12; Set Y has range 30. Which is more spread out overall?
- A** Set X **B** Set Y **C** They are equal **D** Cannot tell
- 11.** Two sets have the same median but different IQRs. They differ in –
- A** center **B** spread of the middle half
C number of values **D** median

SELF-CHECK CODE *find your answer · copy its letter into that problem's box · spell a math word*

21 Z	15 R	4 A	3 E	17 O	Team K Q	9 P	Team J C	7 M	33 X
#1		#2b	#2d	#7b	#7d	#8a	#8b		

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____